

SERVICE MANUAL

Invivo Corporation
3545 SW 47th Avenue
Gainesville, FL 32608

Document TR0529S

Revision 2

GE 3T 32-Channel Cardiac Coil

GE Part Number: M7000LA

MRIDC Part Number: 800529

Manufactured By:



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VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
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WARNING (EN)	This service manual is available in English only. • If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
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FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. • Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése. • Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. • Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
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AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. • Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. • Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. • Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

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BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. • Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. • Neveiciet aprikojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. • Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. • Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. • Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. - Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. - Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. - Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. - Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. - Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. - Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. - Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. - Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. - Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. - Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. - Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. - Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. - Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. - Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. - Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. - No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. - La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . - Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. - Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. - Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. - Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. - Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. - Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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ATTENTION!

A GE field engineer **MUST** perform installation of this coil. Please contact your field engineer to arrange for installation.

Attempts to perform installation of this coil without proper system configuration will result in system errors.

SECTION 1 – INTRODUCTION

1.1 Product Identification and Shipping List

This is a service manual for the GE 3T 32-Channel Cardiac Coil. The 32-channel Cardiac Coil consists of two 16-channel coil halves (anterior & posterior). Each half is uniquely identified by using the labels below:

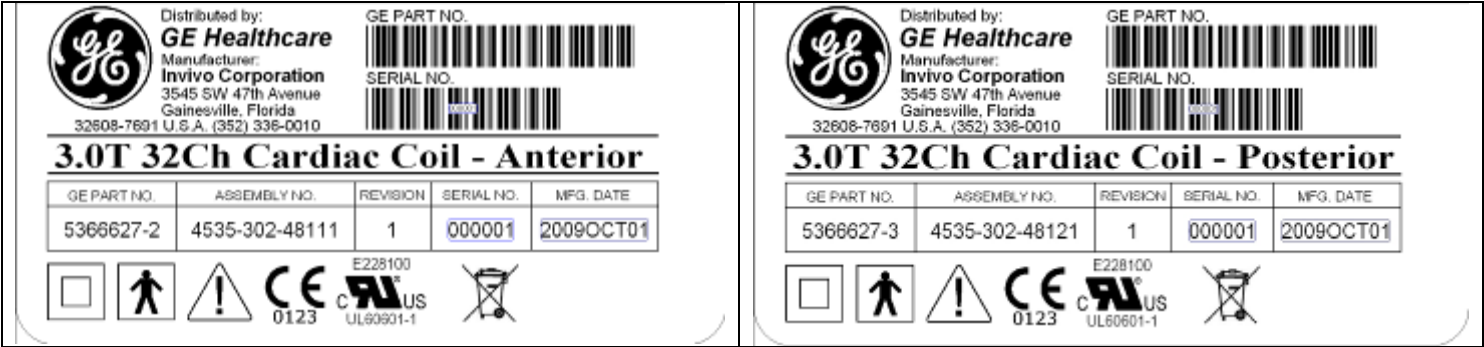


Illustration 1: Cardiac Coil Labels

Illustration 2 shows images of each half of the 3T 32 Channel Cardiac Coil



Illustration 2

1.2 Shipping Contents

SHIPPING LIST BOX 1– TABLE 1-2-1

Description	Invivo Part #	GE Part #	Qty
3T 32Ch Cardiac Coil - Anterior	4535-302-48111	5366627-2	1
3T 32Ch Cardiac Coil - Posterior	4535-302-48121	5366627-3	1
32Ch Cardiac Coil Phantom Positioner (2 pieces)	4535-302-66001	5366627-4	1
32Ch Cardiac Coil - Coil Definition Disk	N/A	5366627-5	1
32Ch Cardiac Coil - Posterior Pad, Bottom	4535-302-58891	N/A	1
32Ch Cardiac Coil - Posterior Pad, Top	4535-302-58901		1
32Ch Cardiac Coil - Head Pad	4535-302-58911	N/A	1
32Ch Cardiac Coil - Foot Pad	4535-302-58921	N/A	1
32Ch Cardiac Coil - Anterior Pad	4535-302-60711	N/A	1
32Ch Cardiac Coil - Operator Manual Insert	4535-302-85131	N/A	1
32Ch Cardiac Coil - Service Manual	TR0529S	N/A	1

1.3 Compatibility

This coil is compatible with the following hardware configurations:

Discovery MR750 3T System

1.4 Related Documentation

- 5500100 Discovery MR750 3T Service Methods
- 1.5T & 3T 32-Channel Cardiac Coil Operator's Manual, 4535-302-58991

1.5 Environmental Requirements

Storage

Store the coil in an air-conditioned scan room or equipment room. To store the coils, a storage space of 48.3 cm/19.0 in (width) x 16.5 cm/6.5 in (depth) x 101.6 cm/40.0 (length) is required.

Dimensions

Coil (with posterior pad):	48.3 cm x 16.5 cm x 101.6 cm	(19.0 in x 6.5 in x 40.0 in)
Phantom Positioner:	50.6 cm x 50.6 cm x 17.7 cm	(20 in x 20 in x 7 in)

Weight

Coil half:	2.95 Kg	(6.5 lb)
Phantom Positioner:	0.68 Kg	(1.5 lb)

SECTION 2 – INSTALLATION PROCEDURE

2.1 Personnel Requirements

Required Persons	Procedure	Finalization
1	15 mins	45 minutes

2.2 Tools and Test Equipment

Item	Qty	Part#
32Ch Cardiac Coil – Coil Definition Disk	1	5366627-5
Phantoms and phantom positioners are needed for the MCQA test.	N/a	See section 3.1

2.3 Safety – Not applicable

2.4 Required Conditions

Condition
The minimum software build to install this coil is 20.0_M4

2.5 Procedure

NOTE: Do not plug in the coil and use AutoCoil to install the coil. This method will not install the coil protocols.

- 2.5.1 Make sure that the latest software build is loaded on the system prior to installing the Cardiac coil. The 32Ch Cardiac protocols will only install if the required conditions are met.
- 2.5.2 Check to confirm if the 32chCardiac coil is already installed on the system software by opening the Service Browser under the Service Desktop Manager. In the Service Browser click on the Configuration tab, then the Coil Database Explorer under the configuration folder on the left.
- 2.5.3 Select “Click here to start this tool”. The Coil DB Explorer window will open and under the "Coil Name" tab there is a "Clinical Coil Name" control. Click on this and there will be a dropdown list of all the logical coil names. There should be three names that start with "C-GE_32Ch Cardiac". This indicates the coil is present in the Coil DB. Reference Illustration 3. If the coil is not present on the list proceed to step 2.5.6.

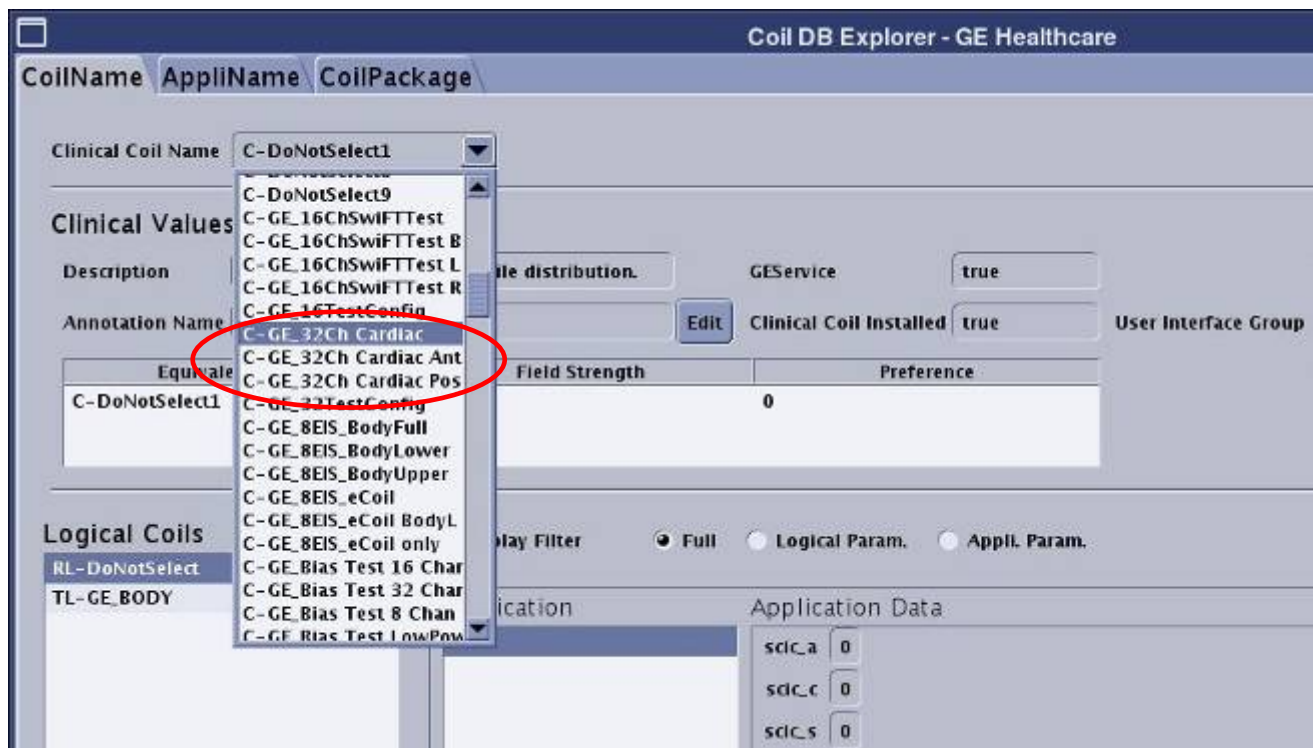


Illustration 3: Confirmation of coil configurations installed

- 2.5.4 The 32Ch Cardiac coil also has two coil protocols that must be on the system. Open the Protocol Management tab. Confirm the coil protocols are installed by viewing the “Chest” protocols under the GE Protocol Library. Two protocols that end in “- 32ch Cardiac” must be present as shown in Illustration 4.

If protocols are not present and the coil CD was just installed one may have to exit the “Chest” protocols by selecting another area such as “Abdomen” and then reselecting “Chest” for the two protocols to appear.

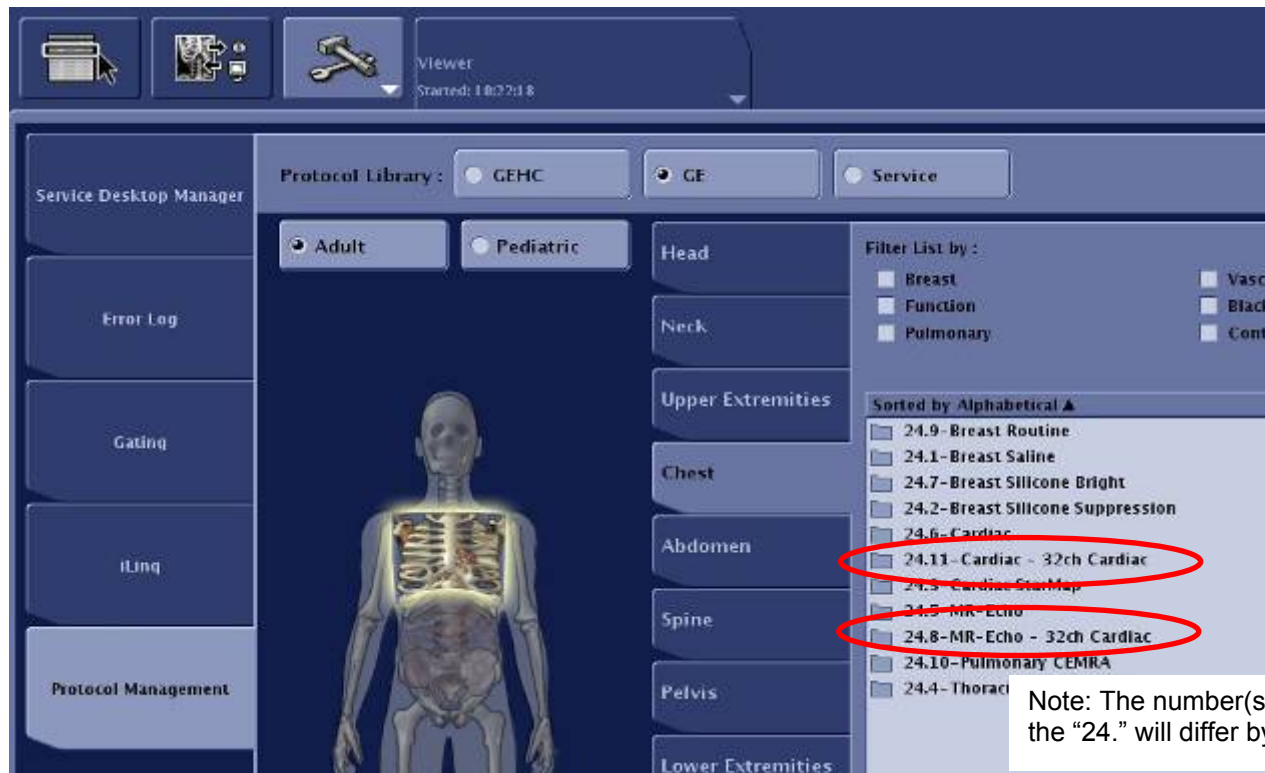


Illustration 4: Confirmation of 32ch Cardiac coil protocols installed

- 2.5.5 If the cardiac coil configurations and protocols are on the system skip to the finalization steps in section 2-6.
- 2.5.6 To install the cardiac coil configurations and protocols, insert the Cardiac Coil Definitions Disc into the default (typically upper) host DVD drive.
- 2.5.7 Open a C-Shell from the Service Desktop manager and type the following commands:

```
su  
<root password>  
mount /mnt/cdrom  
/mnt/cdrom/autorun
```

- 2.5.8 A Coil Data Disk Install window will open. Type “i” to install the CD contents.
- 2.5.9 When the contents of the CD installation is complete, the cursor will appear and you will see the message as shown in Illustration 5. Press enter to exit the Coil CD installation window.

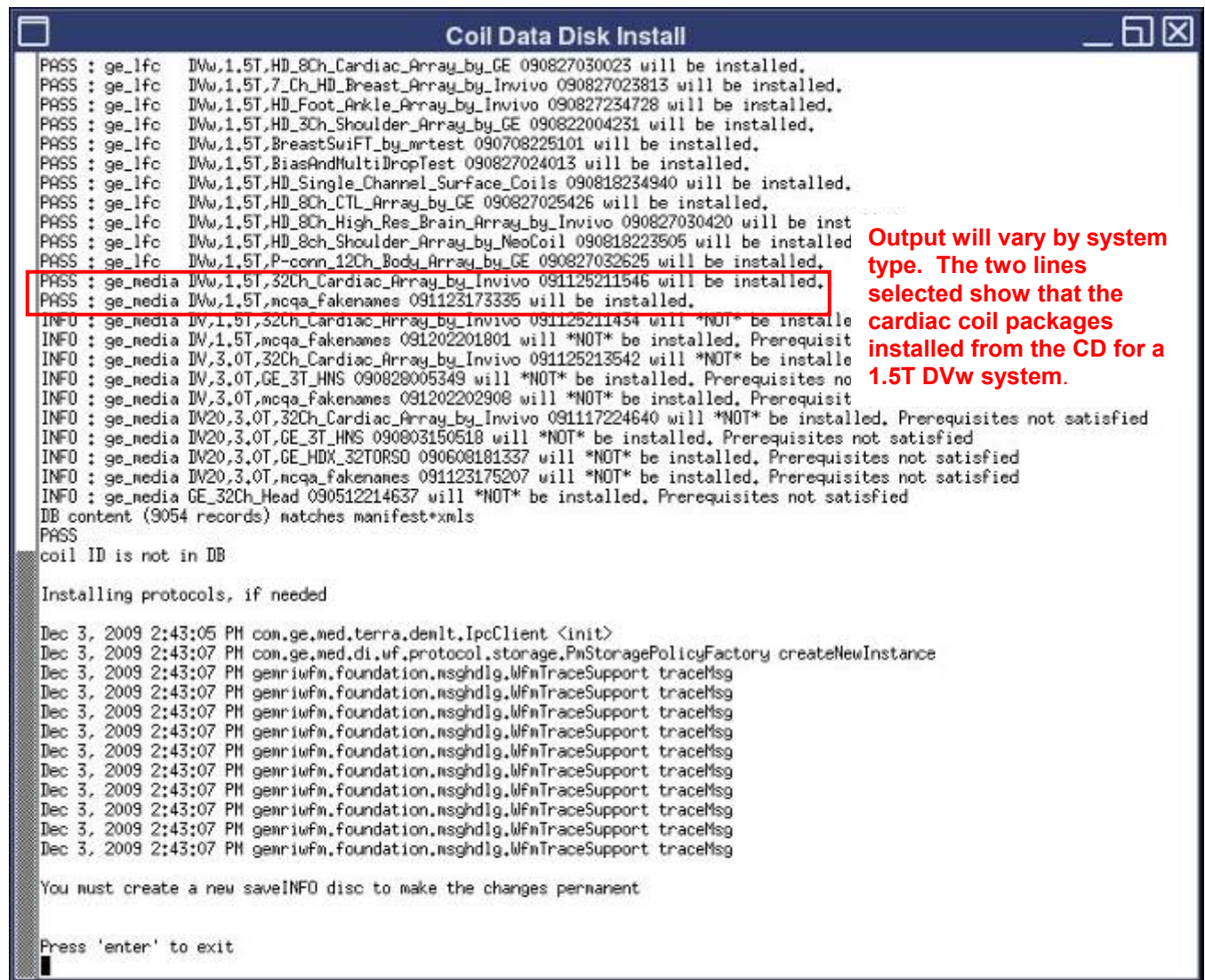


Illustration 5: CD Contents Installation Complete

2.5.10 Type <eject> in the original c shell. Remove the CD from the host DVD drive.

2.5.11 To confirm the coil has been properly installed from the coil CD follow steps 2.5.2 – 2.5.4.

2.6 Finalization

- 2.6.1 Perform a SaveInfo under Guided Install to make the changes permanent.
- 2.6.2 Perform the Multi-Coil Quality Assurance (MCQA) test with this coil. Refer to Section 3.
- 2.6.3 Complete Product Locator information for serialized components. Failure to do this may result in failure of your site to receive future FMI's. Instructions are below.
 - o Go to GIB: http://gib.gehealthcare.com/gib/gib_MainMenu.jsp
 - o After the log in screen, click on [Global Installed Base entries]
 - o Under "Serial Changes to Installed Base", enter the system ID of the system on which this coil is being installed. Click [Submit]
 - o Click on [New Entries]
 - o For both the Anterior and Posterior portions of the coil, enter the Model (Part number), Serial number, Installation Date and "A", then click [Submit].

SECTION 3: 32CH CARDIAC COIL MCQA TEST

Follow this procedure to perform the MCQA test. There are two phantom options that can be used for MCQA test, see section 3.1.

3.1 Tools Required

TOOLS REQUIRED - TABLE 3-1-1

Description	GE Part #	Qty
Legacy: Large Brick Phantom / CTL Phantom / TL Phantoms	5341595 Alternate Part Numbers: 2381682-10 or 2406200 or 5325953-6	1
Or	Or	
Unified TL Phantom Brick	5343347-2	
32Ch Cardiac Coil - Phantom Positioner (2 pieces)	5366627-4	1

3.2 MCQA Setup

The MCQA tool will be run separately for the posterior and the anterior coils. The coil positioning and the phantom setup for the MCQA is the same for both the coil components.

TABLE 3-2-1 Steps for Coil and phantom setup (Anterior or Posterior coil MCQA test)

Visual	Description
	1. Remove any other surface coils (if present) from the cradle.
 Illustration 6	2. Remove any pads present on the coil. The cable must point towards the magnet and plug into ports P1 or P2. Place lower half of phantom positioner on table. Ensure that the positioner is centered in the R/L direction. Insert coil half (anterior or posterior) and place top part of the positioner on the coil as shown in Illustration 6 and Illustration 7.



Illustration 7

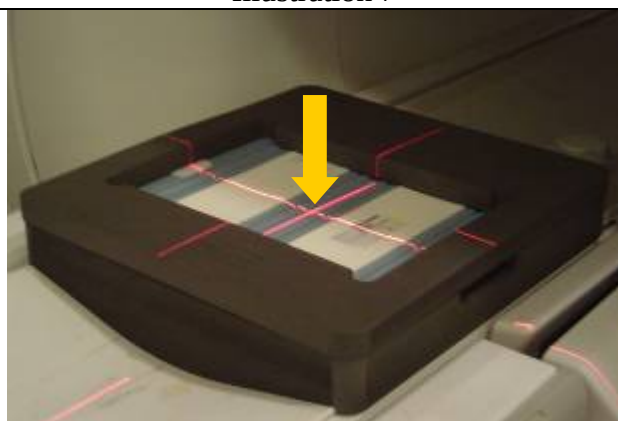


Illustration 8

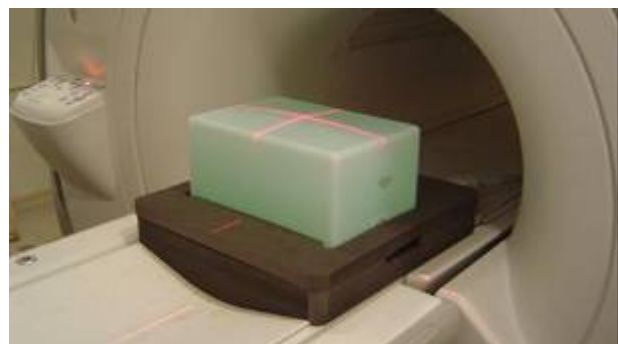


Illustration 9: MCQA SETUP WITH UNIFIED TL BRICK



3. Turn alignment button on

Move table into bore until landmark lasers intersect with crosshairs on the coil as shown in Illustration 8.

Press 'Landmark' button

Place phantom (legacy or unified TL brick) on the coil. Ensure that the phantom is contained within the phantom positioner as shown in Illustration 9 and Illustration 10.

Note: Images show 1.5T phantoms, 3T phantoms contain pink solution.

Press 'Advance to Scan' button.

Illustration 10: MCQA setup with legacy TL Brick

3.3 MCQA Test

3.3.1 At the operator workstation, end the previous exam (if required).

3.3.2 From the Common Service Desktop, select the [Image Quality] tab, choose “Multi-Coil QA Tool”, and then “Click here to start this tool” as shown in Illustration 11.

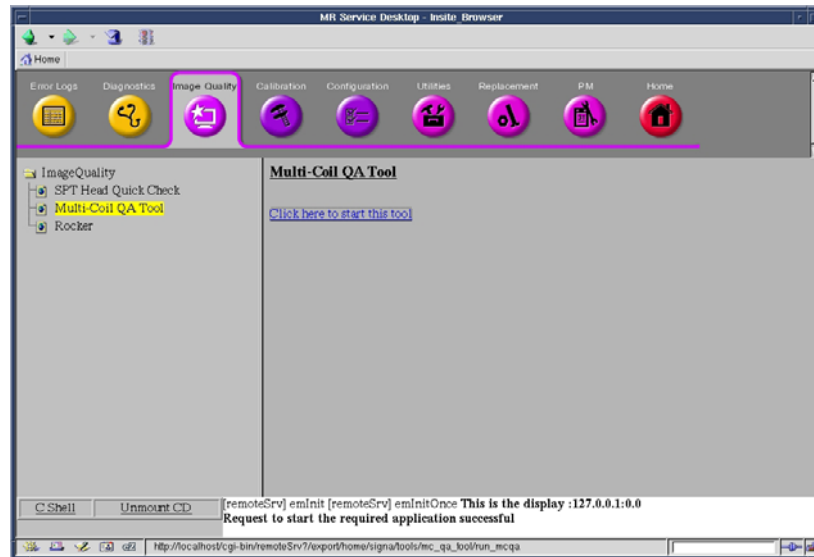
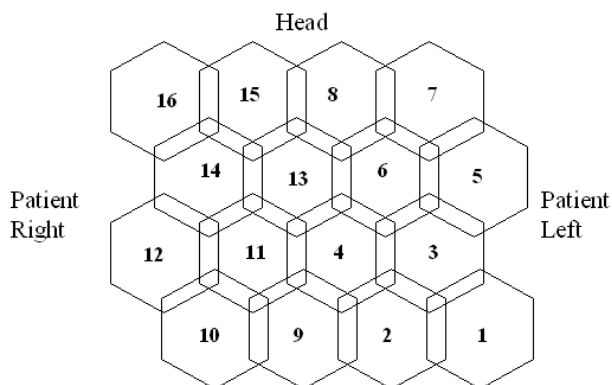


Illustration 11: MCQA from the Service Desktop

3.3.2 Further instructions on using the MCQA tool can be found on the Service Methods CD or the web at: [Troubleshooting -> System -> Multi-Coil Quality Assurance Tool](#).

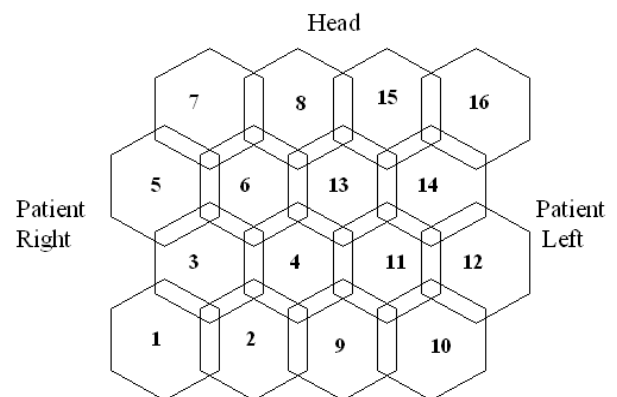
For troubleshooting: The relationship between MCQA image order, coil Receive Pin and coil element are the same (IMG 1 = Rx Pin 1 = Element 1). *Illustration 12* below shows the element layout for the anterior and posterior sections.

Posterior Section Viewed From Top (In positioning pad!)



Foot (cable side, also exits along patient right)
Hexagon diameter = 3.65" measured from center of trace

Anterior Section Viewed From Top



Foot (cable side also exits along patient right)

Illustration 12: Element layout for 32Ch Cardiac Coil

3.4 Periodic Quality Assurance Check

QA checks should be performed during installation and during planned maintenance. Perform the quality assurance checks as outlined below to ensure the coil is operating properly.

3.4.1 Check external cable for cracks or cuts.

3.4.2 Perform Section 3: 32-CHANNEL CARDIAC COIL COIL MCQA TEST

SECTION 4: TROUBLESHOOTING

Note: The Invivo 32-channel cardiac coil is not to be opened for any reason!

4-1 Mechanical Hardware Check

- Verify that there are no loops in the cables.
- Verify that there are no metal or ferromagnetic objects close to the coil, patient or magnet (i.e., safety pin, hair pin).
- Verify that the coil is properly positioned.

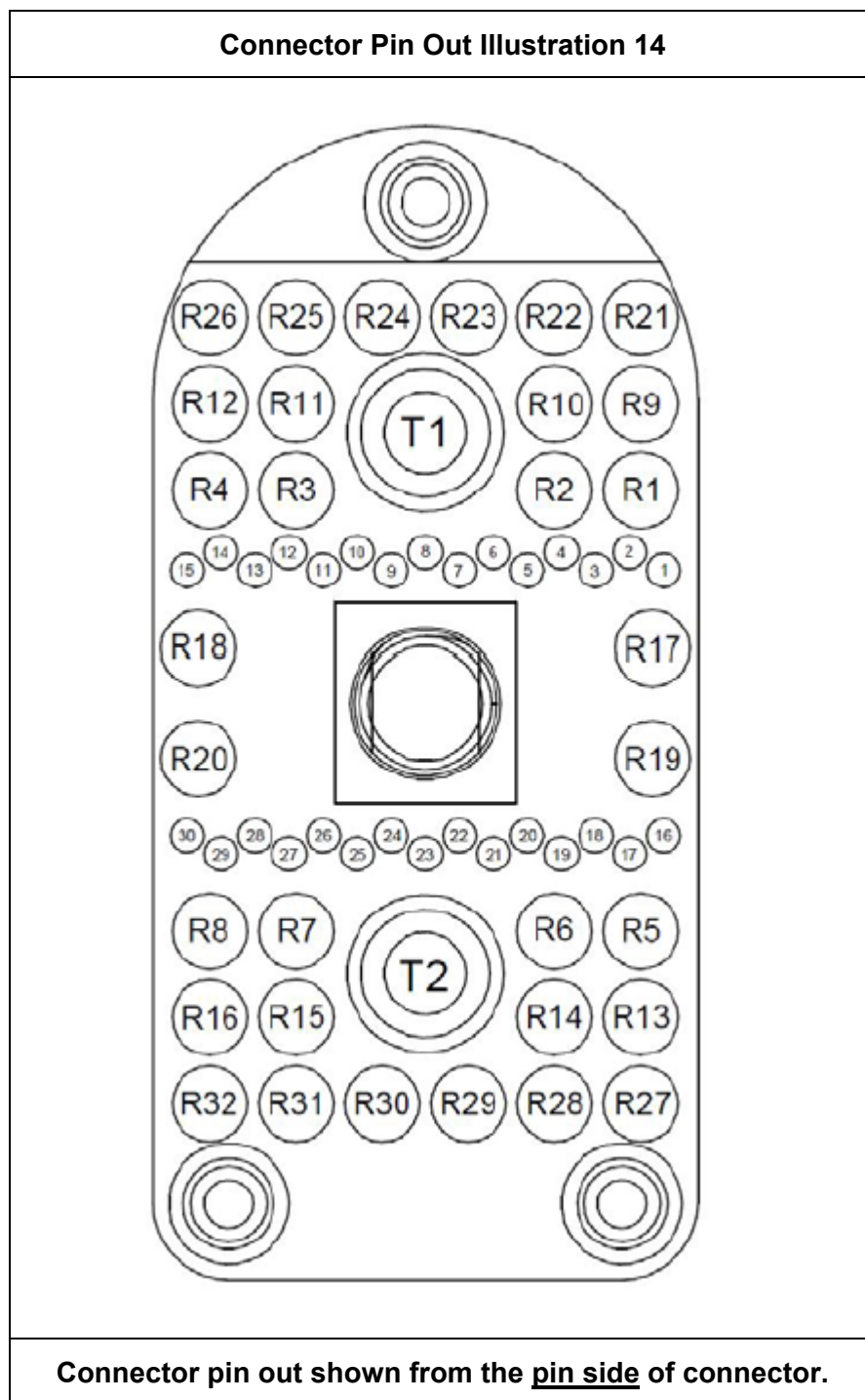
4-2 Electrical Hardware check

Visually inspect all coaxial connectors and pins on the system plug. If there are any broken, deformed or recessed pins or coaxial connectors, replace the coil component.

Coil Connector Illustration 13



View shown from the system side of connector.



NOTE: The center pins of the System Connector *receive coaxial connectors* are extremely fragile [Connector Pin Out Illustration 14, R1-R8]. **Do not attempt to make any measurements at these center pins.**

For GND measurements, use any of the receive coaxial pins' outer shields (R1-R8).

RESISTANCE MEASUREMENTS – TABLE 4-2-1

System Connector		Reading
Positive Lead	Negative Lead	
1	2	< 8Ω
3	4	< 8Ω
5	GND	Open
6	GND	Open
20	GND	340 ±40Ω
21	GND	< 5Ω
22	GND	340 ±40Ω
23	GND	< 8Ω

If one or more of the readings are out of specification, replace the coil. If all of the above readings are correct, redo *Section 3: 32CH CARDIAC COIL MCQA TEST*. If the problem still exists, replace the 3T Cardiac Coil.

4-3 Coil ID Troubleshooting Tips

If the following error is displayed, there may be a problem with the Coil ID coil or system hardware: *The prescribed RF coil and detected RF coil do not match.*

Troubleshooting instructions are available through the Service Methods DVD or online via the path: [Troubleshooting](#) -> [Diagnostics](#) -> [Hardware Location](#) -> [Magnet Room](#) -> [Coil ID Functional Diagnostic](#)

The diagnostic will display the contents of the coil ID chip whenever the connector is inserted. Follow the troubleshooting instructions if the correct ID is not displayed. When the troubleshooting is complete, and if the correct ID is still not displayed, replace the coil.

SECTION 5 – MAINTENANCE

5-1 Coil Care

WARNING!

Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

CAUTION

Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution.

The 3T 32-Channel Cardiac Coil, posterior pads must be cleaned and stored using the following procedure:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- **Do not pour any cleaning solution directly on the coil!**
- Let the coil housing and pads dry before use.
- **Under no circumstances should the coil be placed into any type of sterilizer.**

SECTION 6 – RENEWAL PARTS

6.1 Field Replaceable Units

FIELD REPLACEABLE UNITS LIST – TABLE 6-1

Description	GE Part #	Invivo #
3T 32Ch Cardiac Coil - Anterior	5366627-2	4535-302-86241
3T 32Ch Cardiac Coil - Posterior	5366627-3	4535-302-86251
32Ch Cardiac Coil Phantom Positioner	5366627-4	4535-302-86261

6.2 Other Replaceable Accessories

The following table lists replaceable accessories that may be purchased from Invivo Corporation. The operator manual contains the contact information for Invivo.

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Description	Invivo #
32Ch Cardiac Coil - Head Pad	4535-302-86281
32Ch Cardiac Coil - Foot Pad	4535-302-86291
32Ch Cardiac Coil - Anterior Pad	4535-302-86301
32Ch Cardiac Coil – Posterior Pad Includes: top posterior pad (Invivo part# 4535-302-58901), bottom posterior pad (Invivo part# 4535-302-58891), and straps (Invivo part# 4535-302-59791)	4535-302-86271
Straps Set (Includes both Anterior and Posterior Velcro straps)	4535-302-86311

Revision History

Rev	Date	Author	Primary Reason for Change
A	12-11-09	J. Ciesla	Issued
B	12-16-09	R. David (GE)	Revised related documentation, required conditions, and MCQA sections, changed "array" to "coil"
1	12-17-09	J. Ciesla	Rev 1 Issued
02	1-07-10	J. Ciesla	Corrected label pictures & FRU numbers